Latest version: vuphan314.github.io
This version: 2020/09/03

Vu Hoang Nguyen Phan, M.S.

Curriculum Vitae

vuphan314

Education

2017/08–present **Doctor of Philosophy**, *Computer Science*, Rice University, Houston TX.
- ***Advisor: Moshe Vardi***
- Research: Computational Logic

2017/08–2019/12 **Master of Science**, *Computer Science*, Rice University, Houston TX.
- ***Advisor: Moshe Vardi***
- Research: Computational Logic

2014/08–2017/08 **Dual Bachelor of Science**, *Computer Science & Math*, Texas Tech University, Lubbock TX.

Publication

Except for [Rus18], Vu Phan is the corresponding author.

Thesis

[Pha19] Phan, Vu H. N. "Weighted Model Counting with Algebraic Decision Diagrams". MS thesis. Rice University, 2019. URL: scholarship.rice.edu/handle/1911/107761.

Conference Papers

[Dud20a] Dudek, Jeffrey M. and Phan, Vu H. N. and Vardi, Moshe Y. "DPMC: Weighted Model Counting by Dynamic Programming on Project-Join Trees". In: *Constraint Programming (CP)*. 2020. URL: arxiv.org/abs/2008.08748.

[Dud20b] Dudek, Jeffrey M. and Phan, Vu H. N. and Vardi, Moshe Y. "ADDMC: Weighted Model Counting with Algebraic Decision Diagrams". In: *Association for the Advancement of Artificial Intelligence (AAAI)*. 2020. URL: arxiv.org/abs/1907.05000.

[Rus18] Rushton, Nelson and Phan, Vu H. N. "A Discrete Math Methodology for Computer Science". In: *Frontiers in Education: Computer Science and Computer Engineering (FECS)*. 2018. URL: csce.ucmss.com/cr/books/2018/LFS/CSREA2018/FEC4172.pdf.

Workshop Paper

[Pha18] Phan, Vu H. N. "Syntactic Conditions for Antichain Property in Consistency Restoring Prolog". In: *Answer Set Programming and Other Computing Paradigms (ASPOCP)*. 2018. URL: arxiv.org/abs/1809.09319.

Software Development

2019–2020 **DPMC**, *Dynamic-Programming Model Counter*, `github.com/vardigroup/DPMC`.
- Implemented the model-counting framework in [Dud20a] with Jeffrey Dudek.

Supervisor: Moshe Vardi (Computer Science Department, Rice University).

2018–2020 **ADDMC**, *Algebraic-Decision-Diagram Model Counter*, `github.com/vardigroup/ADDMC`.
- Implemented the model counter in [Dud20b; Pha19].

Supervisor: Moshe Vardi (Computer Science Department, Rice University).

2016–2017 **LED**, *Literate Programming System*, `vuphan314.github.io/LED`.
- Implemented the `Language of Effective Definitions` in [Rus18] by automating its translation into the functional programming language `SequenceL`.
- Implemented an `LED-to-TeX` translator.

Supervisor: Nelson Rushton (Computer Science Department, Texas Tech University).

2015–2016 **L**, *Logic Programming Engine*, `github.com/iensen/LtoASPtranslator`.
- Implemented the language `L` by automating its translation into `Answer Set Prolog`.

Supervisor: Nelson Rushton (Computer Science Department, Texas Tech University).

Computer Skills

Programming Languages

Procedural **Python, C++, Java**.

Functional **Scala, Scheme**.

Logic **SPARC, Clingo, DLV**.

Data **SQL**.

Formatting **TeX, HTML**.

Tools

Linux **Bash, Git**.

Graduate Courses

2020 Fall **Bioinformatics: Sequence Analysis**.

2020 Spring **Statistical Machine Learning**.

2019 Fall **Computer Systems Architecture**.

2019 Spring **Compiler Construction; Multi-Core Computing**.

2018 Fall **Artificial Intelligence**.

2018 Spring **Automated Program Verification; Principles of Programming Languages**.

2017 Fall **Advanced Logic in Computer Science**.

Scholarships

2017–present **Ph.D. Student Tuition & Stipend**, *Computer Science Department*, Rice University.

2016–2017 **General Scholarship**, *Financial Aid Office*, Texas Tech University.

2015–2017 **Academic Achievement Scholarship**, *College of Arts & Sciences*, Texas Tech University.

2015–2017 **Undergraduate Achievement Scholarship**, *Math Department*, Texas Tech University.

2015–2016 **Undergraduate Research Scholarship**, *Math Department*, Texas Tech University.

2014–2016 **Proven Achiever Scholarship**, *Financial Aid Office*, Texas Tech University.